ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B1 · WEATHER AND CLIMATE

Weather words and tools

★☆☆
Grade 4

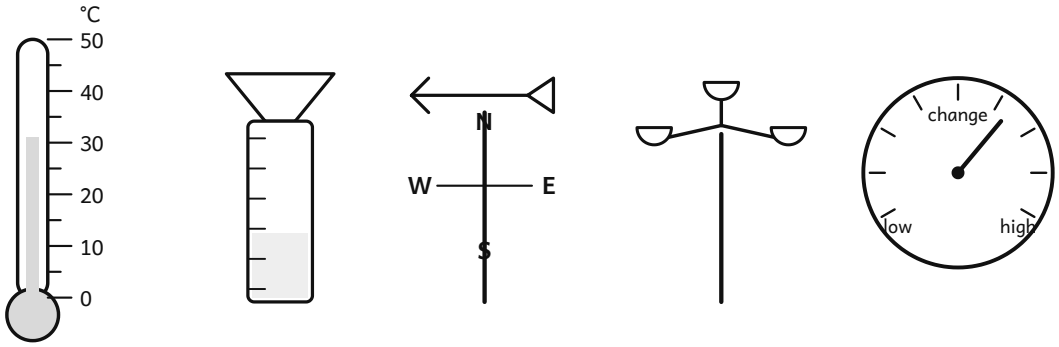
Learning goals

- Tell weather from climate.
- Match instruments to measurements.
- Name cloud types.

1 Weather or climate? Tick one. [4]

Sentence	Weather	Climate
It is very windy this afternoon.	✓	
Sylhet gets a lot of rain every year.		✓
Tomorrow will be foggy.	✓	
Bangladesh has a monsoon season every summer.		✓

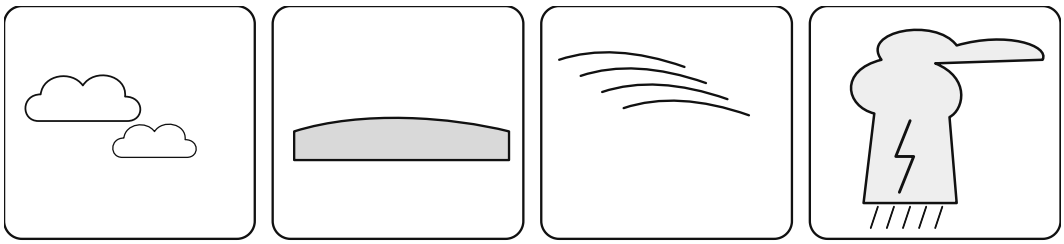
2 Write the name of each instrument and what it measures. [10]



A B C D E

Letter	Instrument	Measures
A	thermometer	temperature (°C)
B	rain gauge	rainfall (mm)
C	wind vane	wind direction
D	anemometer	wind speed
E	barometer	air pressure

3 **Name each cloud type.** [4]



A B C D

A	B	C	D
cumulus	stratus	cirrus	cumulonimbus

4 **Fill in the gaps.** [3]

WORD BANK

sun wind rises shade sinks

Warm air rises. Cool air moves in to take its place. Moving air is called wind. A thermometer should be kept in the shade.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B1 · WEATHER AND CLIMATE

Thinking about air and weather

★★☆
Grade 4

Learning goals

- Predict warm and cool places.
- Explain air movement.
- Use clues to forecast.

1 **Rina measured the temperature in four places at school at 12 noon.** [4]

Place	Temperature (°C)
under a big tree	29
on the tin roof of the shed	45
in the classroom (fan on)	31
on the concrete playground in sun	38

a) Which was the warmest place? Explain why.

The tin roof (45 °C): it is dark metal in direct sunshine and takes in a lot of heat.

b) Which was the coolest place? Explain why.

Under the big tree (29 °C): the leaves shade the ground from the sun.

2 **Why do many old houses in Bangladesh have small openings high up in the walls?** [2]

Warm air rises and escapes through the high openings, and cooler air comes in lower down, keeping the house cool.

3 **The barometer is falling, the wind has changed direction and tall dark clouds are growing. What weather would you forecast? Why?** [3]

A storm with heavy rain and thunder, because falling pressure, changing wind and tall dark (cumulonimbus) clouds are signs of a storm.

4 **Draw a sea breeze. Label the warm land, the cool sea and arrows to show how the air moves.** [3]

Sea breeze

Air rises over warm land, moves out to sea high up, sinks over the cool sea and blows onto the land as a breeze.

ANSWER KEY

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LEARN · PLAY · WONDER

INVESTIGATE · B1 · WEATHER AND CLIMATE

A week at our weather station

★★☆
Grade 4

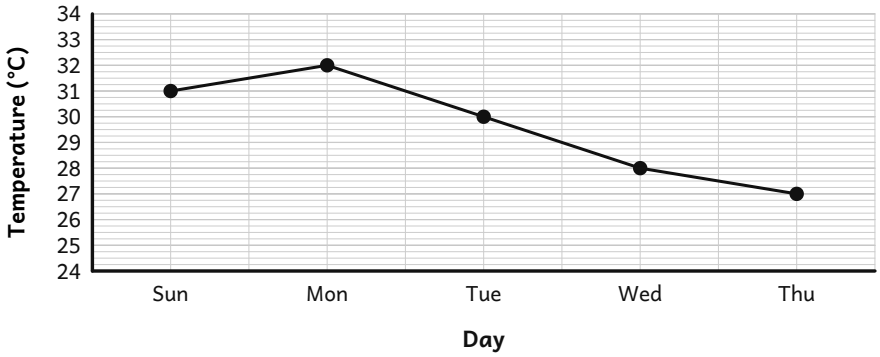
Learning goals

- Record weather data.
- Draw a line graph.
- Make a forecast from trends.

Class 4A recorded the weather at 9 o'clock every morning for one week in June.

Day	Sun	Mon	Tue	Wed	Thu
Temperature (°C)	31	32	30	28	27
Rainfall (mm)	0	2	10	25	30
Wind direction	S	S	SE	SE	SE
Clouds	a few cumulus	more cumulus	grey sheet	dark, tall	dark, tall
Pressure	steady	falling	falling	falling	low

1 **Draw a line graph of the temperature.** [3]



2 **Describe the trend in rainfall during the week.** [2]

Rainfall increased every day, from 0 mm on Sunday to 30 mm on Thursday.

3 **Forecast the weather for Friday. Use at least two pieces of evidence.** [3]

Wet and cool again: rainfall is increasing, the pressure is low, the clouds are dark and tall and the wind comes from the sea (south-east), bringing monsoon rain.

4 **Why did they measure at the same time every day?** [1]

So the results can be compared fairly (temperature changes during the day).

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B1 · WEATHER AND CLIMATE

Harvest before the flood: forecasts for haor farmers

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Grade 4

Learning goals

- See how forecasts help people.
- Use rainfall data.
- Weigh up choices.

Science | Farming | Maths | Technology

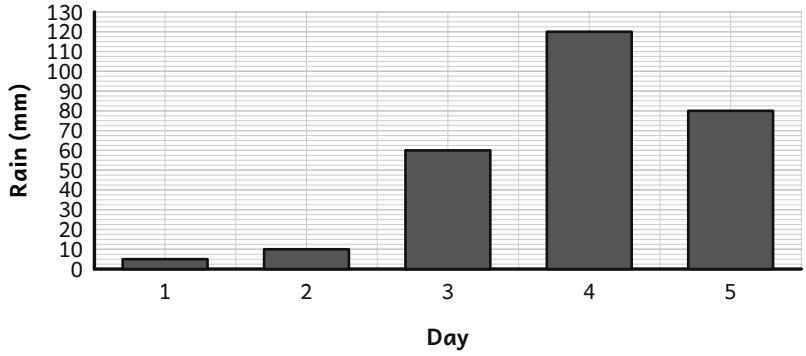
In the **haor** wetlands of the north-east, farmers grow **boro rice** in the dry season. The rice is ready to harvest in **April and May**. But heavy rain in the nearby hills can send a sudden **flash flood** rushing into the haor in just a day or two, drowning the whole crop.

Today, the **Bangladesh Meteorological Department** and the flood forecasting centre send warnings by radio, TV, mobile phone messages and local announcements. If a farmer knows heavy rain is coming, they can harvest early, even if the rice is not completely ripe, and save most of the crop.

Based on information about flash floods in the haors and early-warning services in Bangladesh. Data below are example values.

1 **Forecast rainfall in the hills for the next five days (example).** [3]

Day	1	2	3	4	5
Rain forecast (mm)	5	10	60	120	80



a) Draw the bar chart.

b) On which day should the farmer start harvesting? Explain.

Day 1 or 2: heavy rain starts on day 3 and the flood could come soon after, so harvest before then.

2 **A farmer says, "My rice needs one more week to ripen." What should they do? Give reasons for your answer.** [2]

Harvest now: slightly unripe rice still gives some food and money, but a flash flood would destroy all of it.

3 **Why are mobile phone messages a good way to warn farmers?** [2]

Almost every family has a mobile phone; messages arrive quickly and reach people in remote places.

4 **Design a simple sign that a village could put up to show the flood risk this week (low, medium, high).** [2]

Flood risk sign

e.g. a dial or three flags with pictures (sun / cloud / heavy rain), easy to read for everyone.

ANSWER KEY

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REVIEW AND CHALLENGE · B1 · WEATHER AND CLIMATE

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

M	B	W	K	P	J	K	R	A	I	N
C	L	I	M	A	T	E	N	W	P	G
H	G	F	W	D	M	O	N	E	J	Y
C	I	A	F	O	R	E	C	A	S	T
M	K	M	R	M	C	F	B	T	J	U
D	G	P	O	U	O	H	P	H	E	V
V	A	J	U	N	D	J	T	E	J	U
A	U	D	J	T	S	B	U	R	R	N
N	G	O	J	M	W	O	W	I	N	D
E	E	C	L	O	U	D	O	H	T	A
W	D	E	H	N	A	Y	B	N	G	T

climate cloud forecast gauge monsoon rain vane weather wind

2 Which instrument measures wind speed? [1]

- ○ barometer
- ● **anemometer**
- ○ thermometer
- ○ rain gauge

3 Which cloud is most likely to bring a thunderstorm? [1]

- ○ cirrus
- ○ cumulus
- ● **cumulonimbus**
- ○ stratus

4 Design a simple rain gauge from a plastic bottle. Draw and label it. [3]

My rain gauge

Cut the top off a bottle and turn it upside down as a funnel; mark a ruler scale in mm on the side; put stones in the bottom so it does not blow over; place it in the open.

5 **Challenge. Explain how wind is made. Use the words warm, rises and cool.** [2]

The Sun warms the ground, which warms the air above it. Warm air rises and cooler air moves in to take its place. This moving air is wind.